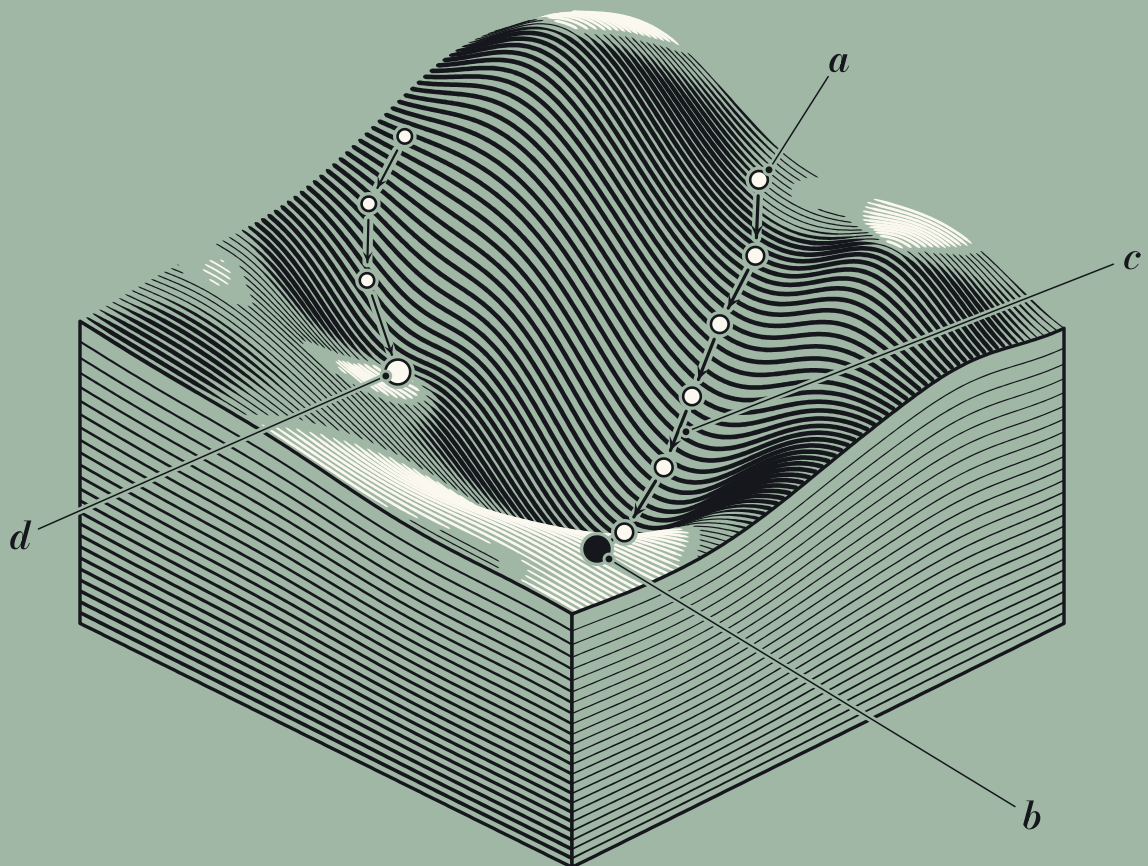


Mathematical Foundations for Machine Learning



*Fig. 1. A loss surface, surveyed and sectioned, with two paths of steepest descent. **a** initial parameters; **b** the minimum; **c** one step along $-\nabla L$; **d** a local minimum, where the second path is caught.*